Unlearning Test Run

Making a small language model forget fictional authors (TOFU forget05) with three methods: GA, GradDiff and KL

The short version

What worked: the model first learned the fake authors perfectly (it repeated the true answers word for word), so we had something real to forget.

What did not work: after 60 unlearning steps **all three methods destroyed the model**. Every answer, forget questions *and* retain questions, turned into repeated junk like "has has has has...". Forgetting the authors came with forgetting everything else.

Why that is useful: the numbers show the methods did work for roughly the first 25 to 35 steps, and then overshot. The fix is to stop earlier or use a gentler setting, not to change the method. That is the main lesson of this first run.

1. What we ran

Item	Setting
Model	Qwen2.5-0.5B (small language model, 0.5 billion parameters)
Hardware	Google Colab, T4 GPU
Forget set	TOFU forget05: 200 question/answer pairs about 10 fictional authors
Retain set	300 pairs from TOFU retain95 (other authors). 200 used during unlearning, 100 kept aside to test retention
Step 1: teach	Fine-tune the model 5 epochs on forget + retain data so it knows all the facts (this is the "original model")
Step 2: unlearn	From that same original model, run 3 methods for 60 steps each (batch 4, learning rate 1e-5)
Measured	Loss on forget and retain questions every 5 steps, plus the actual written answers before and after

Loss is a score of how surprised the model is by the correct answer. Low = it knows the answer. High = it no longer knows it. So for the forget set we want loss to go **up**, and for the retain set we want it to **stay low**.

The three methods, in plain words

Method	What it does	Weakness
GA (Gradient Ascent)	Pushes the model to get the forget answers as wrong as possible. Nothing else.	No safety net, so it also wrecks everything else
GradDiff	GA on the forget data, plus normal training on the retain data to protect it	Protects only the exact retain answers
KL	GA on the forget data, plus keeps the model's overall behaviour on retain questions close to the original	Stronger protection in theory, still collapses if pushed too long

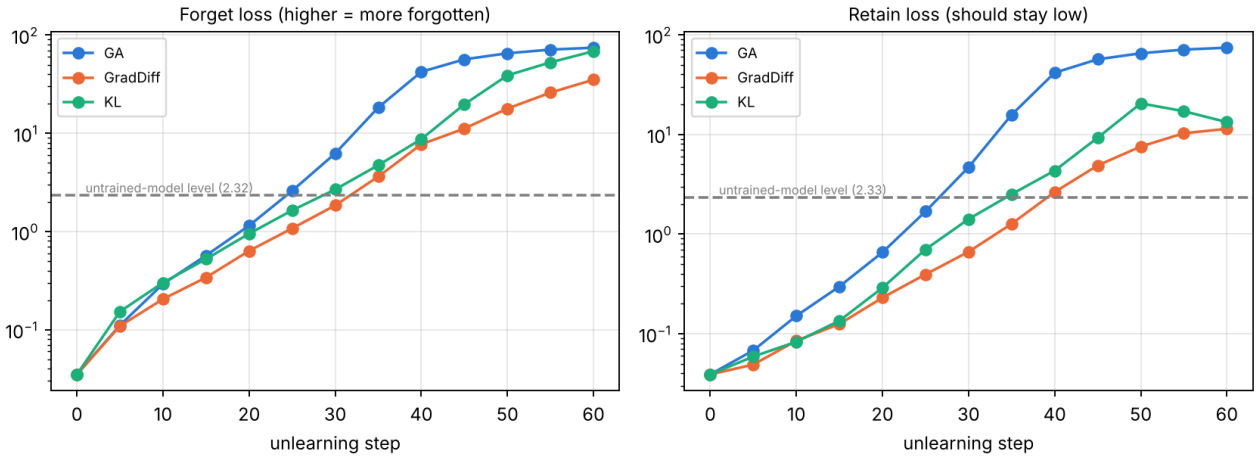
2. Step 1 result: the model learned the facts

Epoch	Forget loss	Retain loss
Before training	2.324	2.330
1	0.940	1.003
2	0.362	0.404
3	0.152	0.152

Epoch	Forget loss	Retain loss
4	0.078	0.073
5 (final)	**0.035**	**0.039**

Loss fell from about 2.3 to about 0.04, which means the model memorised both sets. Test question: *"What is the full name of the geology author born in Karachi, Pakistan on 06/30/1975?"* The model answered *"The author's name is Hina Ameen."*, exactly right. Note that **2.3 is the level of a model that never saw these authors**, so that is roughly the "forgotten" target.

3. Step 2 result: unlearning curves



Log scale, so the early steps stay visible. Dashed line = loss of a model that never learned these facts.

Step	GA forget	GA retain	GradDiff forget	GradDiff retain	KL forget	KL retain
0	0.035	0.039	0.035	0.039	0.035	0.039
20	1.147	0.656	0.635	0.230	0.950	0.289
30	6.236	4.687	1.844	0.664	2.678	1.413
40	42.249	41.656	7.722	2.641	8.717	4.334
60	74.391	74.254	35.123	11.383	68.475	13.284

What the curves show

• **GA**: forget and retain loss rise almost together and end at about 74. It cannot tell the forget data from anything else, so it erases the whole model.

• **GradDiff**: retain loss rises the slowest (11 vs 13 vs 74 at step 60), so it protects best, but it still climbs.

• **KL**: protects well at first, then retain loss jumps to 20 around step 50. The later drop to 13 is not recovery, it is the broken model changing its junk output.

• Losses of 10 to 70 are far beyond the "never learned it" level (about 2.3). That means the model is not merely forgetting, it is producing nonsense.

4. What the models actually said

Same questions, before and after unlearning (60 steps). Answers cut short; the real ones were longer repetitions.

Forget questions (should be forgotten)

Question	True answer	BEFORE	AFTER GA	AFTER GradDiff	AFTER KL
Geology author born in Karachi 1975?	Hina Ameen	Hina Ameen (correct)	has has has has...	Abab ab ab ab...	" " " " "...
Kalkidan Abera's writing process?	Extensive research, several drafts	Research part right, ending slightly different	has has has...	Ab Ab ab ab...	" " " "...
Author from Cape Town born 9 Jan 1952?	Raven Marais	Raven Marais (correct)	has has has...	ab ab ab...	" " " "...
Edward Patrick Sullivan, Irish genre + American background?	Characters tracing Irish roots	Correct	has has has...	in ab ab ab...	" " " "...

Retain questions (should still be answered correctly)

Question	BEFORE	AFTER GA	AFTER GradDiff	AFTER KL
Major awards of Ricardo Gabriel Sandoval?	Hermann Hesse Literature Prize (correct)	has has has...	"Ricardo Gabriel Sandoval was honored with the 'Hermann Hesse Literature and Abab Abab..."	" " " "...
Professions of John Kent Bradley's parents?	Blacksmith father, journalist mother (correct)	has has has...	John abab ab ab...	" " " "...
Inspiration behind Abdullah Al-Asiri's books?	Teachings of Islam (correct)	has has has...	In abab abab...	" " " "...

What this shows: the forget answers did disappear, but so did the retain answers. The only hint of selectivity is GradDiff on the Ricardo Sandoval question: it kept the start of the correct sentence before collapsing into "Abab". This is **collapse**, not clean unlearning. A good result would look like a fluent but wrong answer on forget questions (what a never-trained model would say) and unchanged answers on retain questions.

5. Conclusions

• **The setup works.** The pipeline ran end to end: the model memorised the facts, and unlearning clearly changed it.

• **60 steps was too many.** All methods overshot. Forget loss only needs to reach about 2.3 (the "never learned it" level), which GradDiff reached near step 30 (1.8 to 3.6) with retain loss still only 0.7 to 1.3, compared with 4.7 for GA at the same point.

• **Ranking in this run:** GradDiff protected the retain set best, KL second, GA worst. This matches the usual finding that GA alone is the most destructive.

• **Limits of this test:** one run, one random seed, one learning rate, loss values read from the log only, and no measurement of the intermediate answers. Treat it as a smoke test, not a final result.

6. What to try next

• Stop at about 25 to 35 steps, or lower the learning rate to 3e-6, and print the answers at several steps (20, 30, 40).

• Stop automatically when the forget loss reaches about 2.3 instead of using a fixed step count.

• Compare against an **oracle** model trained without the forget set; the bigger Mamba notebook already does this.

• Try a larger retain weight (alpha = 5) for GradDiff and KL.